

Project Architecture & Technology Stack

BOOK STORE MANAGEMENT SYSTEM

1. Decoupled Architecture Blueprint

The system relies on a clean separation between consumer presentation applications and core backend web servers, communicating via standard JSON API request payloads.

2. Technical Stack Breakdowns

- **Frontend Application Layer:** Built on top of a lightning-fast Vite environment utilizing React 19. Presentation styling is managed with Tailwind CSS. Layout responsiveness is enhanced with Swiper templates, while icons are populated with React Icons.
- **Global State Management:** Redux Toolkit manages synchronous cart mutations across the layout ecosystem, while asynchronous server communications run through configured Axios client interceptors.
- **Backend API Layer:** Formed with Node.js and Express running middleware validation rules to enforce safe CORS boundaries and handle JSON request bodies.
- **Database & Integration Layer:** Leverages Mongoose to model structured collections on top of MongoDB, combined with Firebase SDK elements to drive user profile validation hooks securely.